StepAudio 3: A Voice Model That Thinks While It Speaks, and a Music Model That Plans the Song First

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StepFun's StepAudio 3 family is five AI audio models covering speech, sound effects and music. Its two reports, from September 11 and 12, share one idea: settle the hard part before you hear the output. StepAudio 3 Realtime is a full-duplex speech model, meaning you and it can talk

at once, built on a listen, converse, think and act loop. Think-While-Speaking runs private reasoning alongside speech, so it starts answering while still working a question out, then folds in the tool results. StepAudio 3 Music works the other way: it writes the arrangement out first, in a text shorthand called ABC notation, so harmony and song structure are decided rather than improvised, then renders stereo songs up to 5 minutes 30 seconds long.

That removes the usual tradeoff between a voice model that reasons and one that answers fast. StepFun reports 98.9 on Artificial Analysis' full-duplex bench and 90.6 on MMSU, an audio-understanding test, but the model is absent from that site's main speech leaderboards, so the numbers are hard to check. Its own paper notes gaps in multi-turn constraint following, and on tau-Voice, a customer-service benchmark, it scores 56.0%, just behind Grok Voice Think Fast 2.0's 56.5%. Music is easier to verify: it places fourth on the independent Artificial Analysis music leaderboard, Elo 1092 from 2,043 blind votes, ahead of Suno V5 and Google's Lyria 3 Pro.

Neither ships open weights, and StepFun lists both as free previews with paid versions to come. In both cases the gain comes from an inspectable middle step, a written plan or a private reasoning track, not from one network doing both at once.

Read More: Google's Lyria 3 generates full songs and lets you steer them while they play

Sources:

- StepAudio 3 Realtime Technical Report (arXiv)
- StepAudio 3 Music Technical Report (arXiv)
- StepFun documentation: StepAudio 3 Realtime models and preview access
- StepAudio 3 Music demos and capability walkthrough (StepFun)
- Artificial Analysis music arena vocals leaderboard, blind votes

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